JAN-P-528

26 NOVEMBER 1947

JOINT ARMY-NAVY SPECIFICATION

POWDER, PROPELLENT, M12
(For 20-mm Gun, AN–M2)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective departments and supersedes the following specification:

U. S. Army
50–12–18
30 July 1945

A. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS.

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN–D–98 —Diphenylamine.
JAN–G–155—Graphite (for Use in Ammunition).
JAN–P–193—Potassium Sulfate (for Ordnance Use).
JAN–E–199—Ether, Diethyl.
JAN–D–204—Dinitrotoluene (for Use in Explosives).
JAN–N–244—Nitrocellulose (for Use in Explosives).
JAN–T–458—Tin, Pulverized.
JAN–A–463—Alcohol, Ethyl (for Ordnance Use).

U. S. ARMY SPECIFICATIONS

50–0–1—General Specification for Ammunition Except Small Arms Ammunition.[1]
100–2—Standard Specification for Marking Shipments by Contractors.[1]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[2]

A–2. *Other publications.*—The following publications, of the issue in effect on date of invitation for bids, form a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[2]

INTERSTATE COMMERCE COMMISSION REGULATION

Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

A–3. *Drawings.*—The following drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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76–4–46—Box, Packing, for Metal Liner for Smokeless Powder, Assembly and Details.
76–4–56—Box, Packing, Metal-Wood, M–17, for Smokeless Powder.

B. TYPE.

B–1. This specification covers one type of single perforated, cylindrical grain powder of the following nominal composition:

Ingredient:	*Percent*
Nitrocellulose	97.7
Tin (pulverized)	.75
Potassium sulfate	.75
Diphenylamine	.8
Dinitrotoluene. (Sufficient coating to give desired ballistic properties).	

C. MATERIAL AND WORKMANSHIP.

C–1. *Material.*—The raw material used in the manufacture of propellent powder purchased under this specification shall conform to the applicable specifications listed in paragraph A–1. In addition, the following materials shall be of the type or grade specified as follows:

Nitrocellulose—Joint Army-Navy Specification JAN–N–244, grade C, type I.
Potassium sulfate—Joint Army-Navy Specification JAN–P–193, grade A or grade C.
Ether, diethyl—Joint Army-Navy Specification JAN–E–199, grade A.
Alcohol, ethyl—Joint Army-Navy Specification JAN–A–463, grade 2.
Graphite—Joint Army-Navy Specification JAN–G–155, grade III or grade IV.

C–1a. No reground or reworked powder or nitrocellulose shall be used in the manufacture of powder procured under this specification.

C–2. *Workmanship.*—

C–2a. The process of manufacture used by the contractor shall be approved by the inspector in writing before the contractor proceeds with manufacture. In case any change in the process of manufacture appears to be questionable, the inspector shall bring the matter to the attention of the Chief of the bureau or agency concerned, and notify the contractor of his action in writing within 24 hours.

C–2b. The powder and its standard ingredients shall at all times be protected from the action of direct sunlight and acid fumes.

D. GENERAL REQUIREMENTS.

D–1. With every lot of powder submitted for acceptance, the contractor shall furnish, on official blanks, eight copies of a description sheet giving a complete history of its manufacture.

E. DETAIL REQUIREMENTS.

E–1. *Moisture.*—Maximum, 1.25 percent; minimum, 0.75 percent.
E–2. *Total volatiles.*—Maximum, 2.25 percent.
E–3. *Tin.*—Maximum, 0.95 percent; minimum, 0.75 percent.

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E–4. *Diphenylamine.*—Maximum, 1.0 percent; minimum, 0.70 percent.

E–5. *Potassium sulfate.*—Maximum, 1.00 percent; minimum, 0.50 percent.

E–6. *Ash.*—Maximum, 0.4 percent.

E–7. *Hygroscopicity (percent moisture absorbed between conditions of exposure at 20-percent and 90-percent relative humidity).*—Maximum, 1.5 percent.

E–8. *Dust and foreign material.*—Maximum, 0.075 percent.

E–9. *Stability at 134.5° C.*—When subjected to the test described in paragraph F–4f, the powder shall not change the color of normal methyl violet test paper to a salmon pink color in less than 45 minutes, and shall not explode in less than 5 hours.

E–10. *Ballistic requirements.*—

E–10a. *Velocity.*—The powder under test shall give a corrected velocity of 2,780 f./s. at 90 feet with the 2,000-grain projectile in the velocity and pressure tube (Mk. 2—British chamber).

E–10b. *Velocity uniformity.*—The maximum dispersion in velocity shall be not greater than 60 f./s. and the average deviation from the mean shall be not greater than 20 f./s.

E–10c. *Pressure.*—The average pressure at a weight of charge to give a velocity of 2,780 f./s. at 90 feet (corrected) with the 2,000-grain projectile shall not exceed 48,000 p. s. i. (corrected).

E–10d. *Maximum weight of charge.*—The charge required to impart the prescribed velocity to any projectile shall not exceed 505 grains.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. *Size of lot.*—Maximum, 65,000 pounds; minimum, 50,000 pounds.

F–2. *Sampling.*—

F–2a. *Ballistic sample.*—The inspector shall select 10 boxes at random and shall withdraw from each of the 10 boxes a ballistic sample consisting of 2 pounds. These samples shall be placed in appropriate glass bottles with rubber stoppers (or other container approved by the contracting officer), sealed and labeled. All labels shall state the lot number, date of manufacture, date of packing, name of plant, name of manufacturer, total weight of the lot as packed, and the number of the box from which the sample was taken. In addition, the bottle shall be marked "Ballistic sample." These samples shall then be sent to the place specified by the contracting officer. Upon conclusion of the ballistic tests a 2-pound composite sample of the powder lot shall be retained at the ballistic range performing the acceptance test for a period of 6 months thereafter. This sample is for use in subsequent charge establishment with different weight of projectile, if necessary.

F–2b. *Chemical sample.*—From each of the containers used for the ballistic samples, withdraw approximately a 1½-pound sample for the chemical tests. Portions of these samples shall be reserved for the 134.5° C. stability test described in paragraph F–4f, and the remainder of the portions blended into a composite sample of approximately 10 pounds for chemical and surveillance tests.

F–3. *Inspection.*—

F–3a. *Army.*—Inspection shall be made in accordance with U. S. Army Specification 50–0–1. Final inspection shall be made on each lot of powder after it is blended and packed, and shall be at the place of manufacture unless otherwise specified in the contract or order.

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F–3b. *Navy.*—Inspection shall be made at the place of manufacture.

F–4. *Tests.*—Tests shall be made in accordance with the following paragraphs. For Navy purchases, the tests shall be made at a Government laboratory unless otherwise specified in the contract or order.

F–4a. *Moisture.*—Ten gm. of powder shall be accurately weighed in a ground-glass-stoppered weighing bottle. The bottle shall be 60-mm. diameter by 30-mm. deep. The unstoppered bottle and the contained powder shall be heated in a vacuum oven (not less than 25-inch vacuum) for a period of 6 hours at a temperature maintained at 55° ±2° C. The stoppered weighing bottle and contents, after being cooled in a desiccator for ½ hour, shall then be weighed and the loss in weight shall be calculated as percent moisture.

F–4b. *Total volatiles.*—

F–4b (1). *Preparation for test.*—

F–4b (1) a. *Dibutylphthalate reagent.*—Prepare a large quantity of dibutylphthalate reagent containing approximately 1 gm. of diphenylamine per liter as follows: Dissolve 1 gm. of diphenylamine in 100 ml. of hot dibutylphthalate and shake vigorously. Add dibutylphthalate to make 1 liter. Shake vigorously. (To make more or less reagent change the amount of the chemicals proportionately.) When tested separately in the solution-evacuation apparatus it should lose less than 10 mg. It may be necessary to pretreat the dibutylphthalate as follows, to meet this requirement: Heat the dibutylphthalate for 2 hours at 145° to 150° C. while bubbling dry air through it.

F–4b (1) b. *Preparation of solution tubes.*—Place 10 clean, dry steel balls, $\frac{1}{16}$ inch in diameter and 50 ml. of dibutylphthalate reagent into each of a series of clean, dry solution tubes. (See fig. 1.) The dibutylphthalate reagent may be added through a small funnel either by means of a graduated cylinder or from an automatic 50-ml. delivery device provided with a larger-than-normal outlet. Clean the inlet of each solution tube with absorbent cotton held by steel forceps. Using the forceps, insert a wad of pyrex glass wool, or equivalent (about 0.2 gm.) in the inlet of each tube at such a point that it will not be in danger of falling into the tube and at the same time will not be in too close contact with the cork stopper which is to be subsequently inserted.

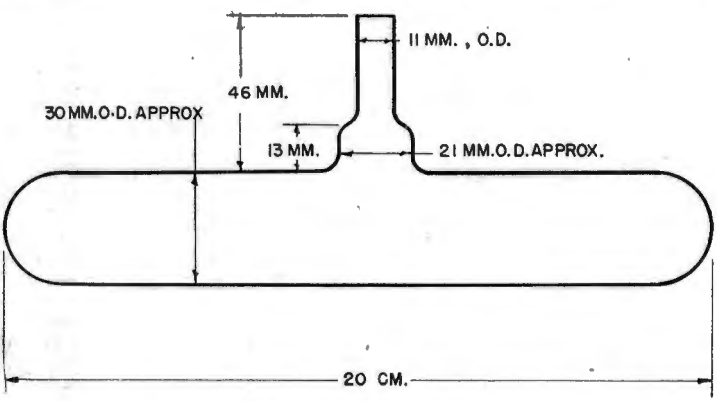


FIGURE 1.—Solution tube.

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By means of 5-cm. pieces of rubber tubing [3] attach each solution tube to the manifold of a rocking device inside an oven [4] maintained at 85° C, Attach to the manifold a vacuum line assembly.[5] (See fig. 2) and a vacuum pump which is capable of consistently maintaining a pressure of 1 mm. or less.

Close stopcock G. With stopcock A open and stopcock B and needle valve E closed, turn on the rocking motor and vacuum pump, and evacuate the tubes to a pressure of 1mm. or less as indicated by the McLeod gage. This pretreatment procedure is accomplished very easily because the evacuation is started when the dibutylphthalate reagent is still at approximately room temperature. The low boiling impurities are removed gradually at 1 mm. pressure as the temperature rises. At the end of 1½ hours, stop the vacuum pump and rocking motor, open stopcock B cautiously and admit dry air slowly by gradually opening the needle valve E. If air is admitted too rapidly, the wads of pyrex wool, or equivalent, may be drawn into the solution tubes. Wearing a pair of gloves, remove the solution tubes from the oven, leaving the 5-cm. rubber connecting tubes attached, and place the solution tubes in a wire-screen tray. (See fig. 3.)

Immediately insert small protective drying tubes [6] (see fig. 4), and allow the tubes and contents to cool to room temperature. This will require from 30 to 45 minutes. The small drying tubes will prevent the admission of atomspheric moisture during the cooling period.

F–4b (1) c. *Determination of loss of weight of solution tubes.*—The determination of the loss in weight of a solution tube involves the measurement of a decrease of 20 to 40 mg. (for powder containing 1 to 2 percent total volatiles) in the weight of a 170-gm. system. Therefore,

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[3] Rubber pressure tubing should have a 5/16-inch bore, 3/16-inch wall. Use 5-cm. pieces for connecting the glass tubes to the copper manifold. Cut off a number of pieces, boil these in 5 percent sodium hydroxide solution for 10 to 15 minutes, wash thoroughly and dry at 100° C. The rubber tubing should be examined after each volatile solvent determination and replaced when necessary.

[4] The oven should be fitted with a rocking device and electric motor. By means of the rocking device the tubes are tilted back and forth through an angle of approximately 90° at the rate of 12 complete cycles per minute. The oven designated as Catalog No. 95105–A by the Central Scientific Co. has been found satisfactory and should be provided with a safety latch to replace the regular latch. A steam oven, if available, may be used in place of an electric oven.

[5] Vacuum line assembly should be as shown on figure 2. The air and vapors from the solution tubes are passed through a T-tube leading to a McLeod gage, a T-tube leading through an aspirator bottle I, filled with Drierite, to a needle valve E, a three-way stopcock F, a 500-ml. absorption bottle H containing silica gel, a trap D, a T-tube leading to a stopcock G, and a pressure regulator C. The bottle of Drierite I serves to dry the air which is admitted to the tubes in the oven. The needle valve E permits a much closer control of the pressure during the initial stage of the evacuation than can be obtained by stopcock B. The three-way stopcock F serves to admit nitrogen to the tubes in the oven. The absorption bottle of silica gel H is included to absorb the volatiles from the powder and thus make it unnecessary to change the oil in the vacuum pump frequently. The silica gel may become ineffective as shown by the inability of the vacuum pump to exhaust the system to 1 mm., or less. If this happens, the silica gel should be replaced with fresh material. The trap D serves to keep dibutylphthalate reagent, in the pressure regulator, away from absorption bottle H in case stopcock A is accidentally closed when the vacuum pump is stopped. The stopcock G serves to admit air into the vacuum pump and pressure regulator after the tubes have been filled with nitrogen. The regulator C, containing dibutylthalate reagent serves to maintain the desired interior pressure.

[6] Sixteen small protective drying tubes are required. These tubes contain indicating drierite (or equivalent). Store them in a desiccator when not in use. Keep the drierite anhydrous (blue) by occasionally putting the tubes in an oven at 130° C. overnight.

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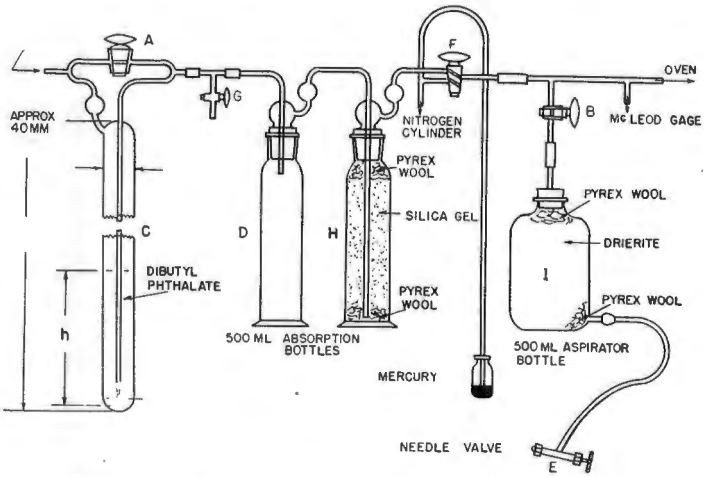


FIGURE 2.—Vacuum line assembly.

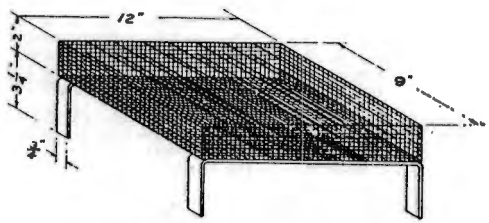


FIGURE 3.—Wire screen tray.

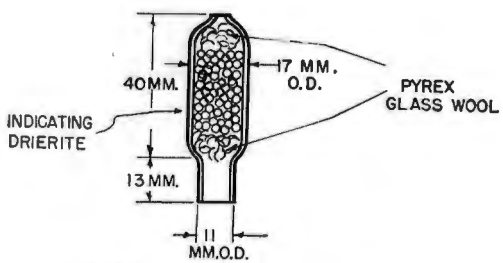


FIGURE 4.—Small protective drying tube.

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the weighing and conditioning procedure is very important. After the tubes have cooled to room temperature, keep them stoppered, except while being weighed, to prevent absorption of atmospheric moisture by the very dry dibutylphthalate reagent. Remove the rubber connectors and protective drying tubes from all of the solution tubes

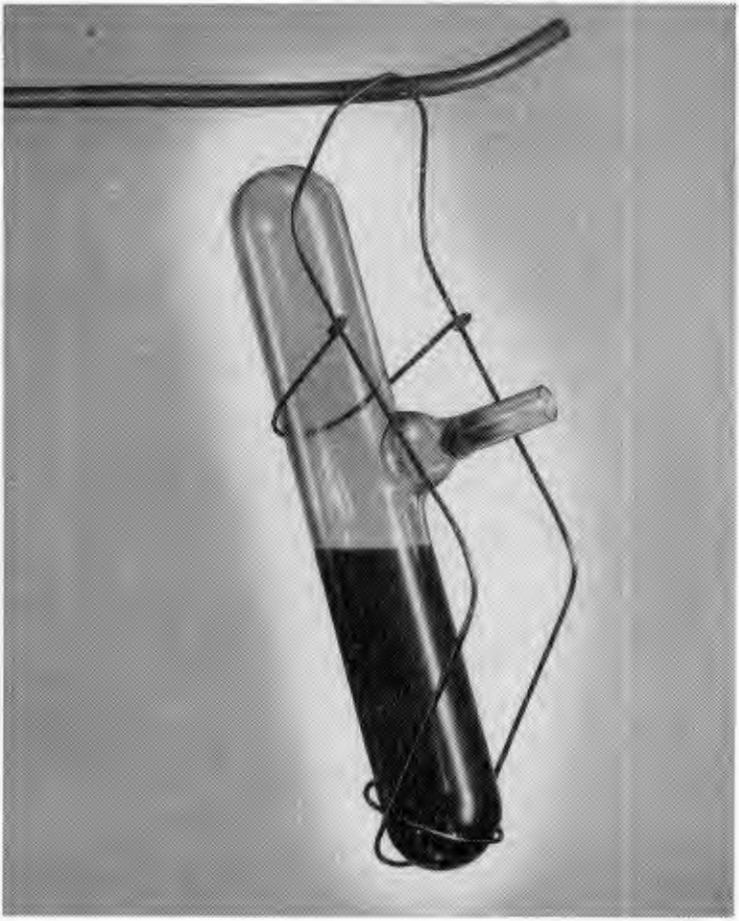


FIGURE 5.—Wire tube holder.

and insert No. 1 cork stoppers. Wipe the tubes with a wet towel and then wipe (but do not rub) with a clean dry cloth which is substantially free from lint. This operation is for the purpose of cleaning the tubes, removing electrostatic charges, and hastening the attainment of equilibrium with moisture content of the air. Place the solution tubes

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in a wire-screen tray, cover with a cloth, and allow to stand near the balance for at least 30 minutes. At no time after a tube has been conditioned should it be touched by the fingers. In putting a tube on the balance to weigh, handle it only by its inlet tube after covering the latter with a small piece of tissue paper. Place each solution tube in a wire tube holder and weigh (see fig. 5).

F–4b (2). *Test procedure.*—

F–4b (2) a. Take approximately 2 gm. of whole grains of powder for each determination. Weigh the samples to the nearest 0.01 gm. in an aluminum weighing dish, place in a series of small weighing bottles, and stopper the bottles immediately. Weigh 2 or 3 of the tubes accurately for use in making blank determinations; that is, determination of the small change in weight that occurs when the tubes of dibutylphthalate reagent are heated and evacuated under the same conditions that prevail in the analysis, making these weighings, as well as all other accurate ones, by using a counterpoise [7] (see fig. 6) on the right-hand side of the balance. Keep this counterpoise near the bal-

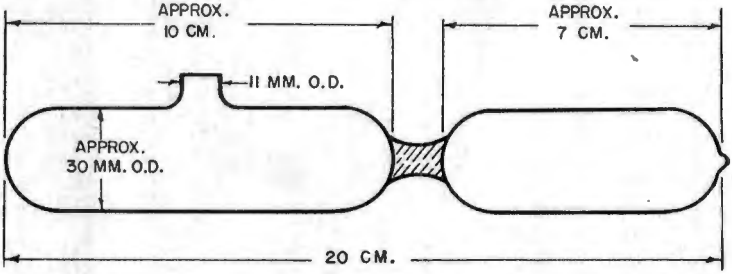


FIGURE 6.—Counterpoise.

ance so that it will be practically in equilibrium with the prevailing conditions of temperature, pressure, and humidity. Do not wipe the counterpoise when the solution tubes are wiped. Make at least 2 blank determinations (the first and eighth tubes) if a total of 8 tubes is being used, but make at least 3 blank determinations if a total of 16 tubes is being used (the first, eighth, and sixteenth tubes). Add the weighed samples to the tubes and weigh the tubes accurately.

F–4b (2) a (1). From each tube remove the cork stopper, remove the wad of pyrex wool with forceps, add the sample with the aid of a small metal funnel having an outlet of approximately 9 mm. outside diameter, reinsert the wad of pyrex wool or equivalent, weigh the tube accurately, shake it if necessary to loosen any grains which may be adhering to the walls of the tube, connect it to the manifold in the electric oven, and rock the tube mechanically. It is considered desirable, although not essential, to have the samples dissolve completely during the overnight rocking period at 85° C., and they are much more likely to dissolve if the steel balls and grains of powder all move freely in the

[7] A counterpoise tube is a glass tube containing a suitable number of steel balls so that it will have approximately the same weight, volume, and exterior surface as a solution tube containing 50 ml. of dibutylphthalate. This tube should be kept standing near the balance, covered with a cloth to protect it from dust.

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tubes when they are put in the oven. If the samples are of such a nature that they do not adhere firmly to the walls of the tubes on standing in the dibutylphthalate solutions for 30 minutes (this must be determined by experience), the procedure just described may be shortened in two respects: (1) the weighed samples may be transferred directly from the aluminum weighing dish to the solution tubes and (2) the whole series of tubes may be put into the oven at the same time. In some cases it has been found advantageous to rock the weighed solution tubes in a cradle from the time they are weighed so that all can be put into the oven at the same time.

F–4b (2) b. As soon as all the tubes have been placed in the oven, turn on the rocking motor. Open the nitrogen cylinder valve to allow a slow stream of nitrogen to escape through the mercury trap. With stopcock A open and stopcock B closed, evacuate the tubes, fill them with nitrogen, evacuate a second time, and again fill with nitrogen. This is accomplished very easily by manipulating stopcock F. The nitrogen should be admitted rather slowly to prevent the gas stream carrying any particles of foreign matter from the rubber tubing or manifold into the tubes. Leave stopcock F turned to connect the oven with the tube leading to the nitrogen overflow bottle containing the mercury. Then admit air to the pressure regulator, vacuum pump, etc., by opening stopcock G. The operation of filling the tubes with nitrogen is facilitated by having them at approximately room temperature at this point so that they can be evacuated quickly without taking any special precautions. Rock the tubes at a temperature of 85° C. for approximately 15 hours. Ordinarily the samples dissolve in the dibutylphthalate reagent under these conditions. It is essential that the samples be softened by dibutylphthalate reagent but not essential that they dissolve completely. After the tubes have been rocked for approximately 15 hours, remove the volatile matter by evacuation at 5 mm. pressure for 2 hours. The beginning of the evacuation, while most of the volatile matter is being removed, is the critical point of the determination. Assuming that the tubes are properly conditioned and weighed, the accuracy of the results depends upon the care with which this operation is performed. With the rocking device in operation and with stopcocks A and B and the needle valve E open, turn on the vacuum pump. Evacuate at gradually decreasing pressures by slowly closing needle valve E until a pressure of 5 mm. is reached. The lowering of the pressure from about 50 mm. to 5 mm. must be done cautiously over a period of approximately 10 minutes while watching the operation through the glass door of the oven. Close stopcocks A and B, and continue the evacuation for 2 hours at 85° C. and 5 mm. The pressure must be reduced so gradually that bubbles of volatile matter leave the dibutylphthalate solution without danger of causing mechanical loss. At the end of 2 hours, stop the vacuum pump and rocking motor. Open stopcocks A and B and gradually admit air into the system through needle valve E. If stopcock A is not opened at this point, the dibutylphthalate solution will flow out of the pressure regulator. Remove the tubes from the oven, insert the protective drying tubes in the rubber connectors, let the tubes cool to room temperature and insert cork stoppers. Condition and weigh the tubes in the same manner as that previously described. If a special ink bas been used instead of hydrofluoric acid solution to put numbers on the tubes, care must be taken to avoid rubbing off any of the ink. As a matter of good practice and

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in order to obtain minimum blank values, the conditioning procedures before and after the removal of the volatile matter should be similar.

$$\text{Percent total volatiles} = \frac{100\ (A + B)}{W}$$

where

A=decrease in weight of sample tube.
B=average change in weight of blank tubes.
W=gm. of sample.

The changes in weight of the blank tubes are usually negative, but may be positive (possibly due to humidity changes): They seldom exceed 3 mg. In all cases the average change in weight, B, is algebraically added to A; that is, if B is negative it is subtracted from A, if positive it is added. Blank values are usually concordant within 1 mg. If they are not concordant within 2 mg., the analysis should be repeated. This method has a precision of 0.10 percent, that is, results should agree with 0.01 unit when expressed as percent total volatiles. If they do not agree this closely, the analysis should be repeated.

F–4c. *Hygroscopicity.*—When, in the opinion of the inspector, it is considered necessary to perform a test for the determination of hygroscopicity, the following procedure shall be used:

Ten to 15 gm. of powder, weighed accurately, is spread on 3½-inch watch glass and placed in a desiccator containing a sulfuric acid-water solution of such concentration as to give 20 percent humidity (59.1 percent H_2SO_4). The desiccator is placed in a constant temperature oven at 30° C. until the weight becomes constant. The sample is then placed in a desiccator containing sulfuric acid of such concentration as to give 90 percent humidity (18.6 percent H_2SO_4) and the desiccator is placed in the constant temperature oven at 30° C. until the weight becomes constant. The hygroscopicity or percent of moisture absorbed between the conditions of exposure at 20 percent and 90 percent relative humidity is calculated as follows:

$$\text{Percent hygroscopic moisture} = \frac{C-B}{W} \times 100$$

where

W=weight of sample.
B=constant weight at 20 percent relative humidity.
C=constant weight at 90 percent relative humidity.

F–4d. *Dust and foreign material.*—Place approximately 100 gm. of the powder in a tared glass or aluminum dish, dry in a desiccator over 95–96 percent sulfuric acid for 18 to 20 hours, and weigh. Pass the powder through a separator (see fig. 7) in which the dust is separated from the powder grains by means of a stream of air of suitable velocity. Weigh the powder and again pass through the separator; the operation shall be repeated until constant weight is attained. Calculate the loss in weight of the powder as percent dust and foreign matter.

F–4e. *Composition.*—

F–4e (1). *Ether extractive material.*—Extract a weighed portion of approximately 5 gm. of the powder with diethyl ether containing not more than 4-percent alcohol for a period of not less than 16 hours using a Knorr or other suitable extraction apparatus until the extraction is complete. The extract contains diphenylamine stabilizer and dinitrotoluene coating material which shall be determined in accordance with paragraph F–4e(1)a and F–4e(1)b respectively. Evaporate the extract to dryness with a current of air and dry to constant weight in a vacuum desiccator over calcium chloride.

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F–4e (1) a. *Diphenylamine.*—Dissolve the residue in the flask from the ether extractive determination in 100 ml. of ether. Transfer the ethereal solution to a 250-ml. beaker and add 0.6 ml. of bromine carefully. Place the beaker on a steam bath, maintained at approximately 75° C., and evaporate the ether with a current of dry air. Add 40 ml. of approximately 95-percent (by volume) ethyl alcohol solution and heat the liquid to boiling. Add 40 ml. of water at room temperature and allow the beaker to remain on the steam bath for 10 minutes. Filter the hot solution through a tared filtering crucible with a mat which will permit rapid filtration and wash the retained precipitate of tetrabromdiphenylamine with 80 ml. of boiling approximately 47.5-percent (by volume) ethyl alcohol solution and finally with hot water. (The alcohol solution serves to dissolve the dinitrotoluene, which remains in solution when the alcohol is diluted to approximately 47.5-percent by volume. This strength alcohol solution does not have appreciable solvent action on the tetrabromdiphenylamine.) Dry the crucible and contents for 1 hour at 100° C., cool in a desiccator and weigh. Calculate the increase in weight to percent diphenylamine in the powder on a total volatiles-free basis.

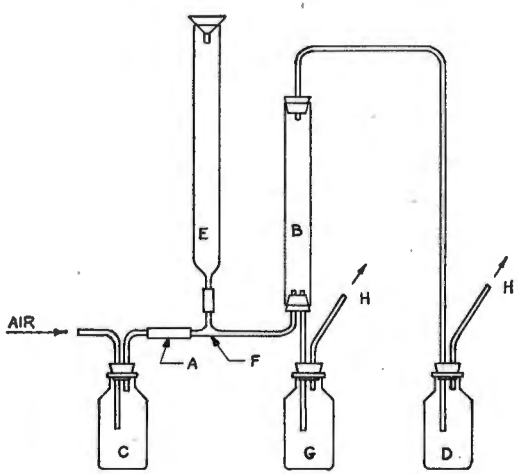


FIGURE 7.—Dust removing apparatus.

A—Copper T-tube.
F—Copper nozzle, outlet not more than 1.5-mm. in diameter and placed not more than 2-mm. from the opening of the T-tube.
E—Charging tube.
B—Separating tube.
C, D, G—8 oz. wide mouth bottle.
H—Safety tubes, cotton gauze over end of tube.

OPERATIONS: The air stream emerging from the jet (A) produces a venturi effect, drawing the sample from the charging chamber (E) and expelling it into the large upright separator (B) where the grains are separated from the dust by virtue of their larger size, the dust being carried over into collector (D) while the whole grains, under the influence of gravity, fall back to collector (G). The cotton gauze is put on top of the safety tube (H) to insure that no grains can escape the apparatus during operation.

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$$\text{Percent diphenylamine} = \frac{34.87A}{W}$$

where

A=gm. of tetrabromdiphenylamine.
W=gm. of sample on a total volatiles-free basis.

F–4e (1) b. *Dinitrotoluene.*—Calculate the percent dinitrotoluene as follows:

$$\text{Percent dinitrotoluene} = \frac{100\ (B - 0.01DW)}{W}$$

where

B=gm. of ether extract (par. F–4e (1)).
D=percent diphenylamine (par. F–4e (1) a).
W=gm. of sample (volatiles-free).

F–4e (2). *Potassium sulfate.*—

F–4e (2) a. *Preparation of reagent.*—Dissolve 100 gm. of chemically pure trisodium cobaltinitrite $Na_3Co\ (NO_2)_6\ .H_2O$ in water and dilute the solution to 500 ml. Store the solution in a dark bottle and avoid direct sunlight. This reagent is stable for only one week. Reagents older than one week should be discarded, as they tend to give high results. Standardize each lot of trisodium cobaltinitrite against a known solution of K_2SO_4 containing 0.050 gm. per 25 ml. In the standardization, use a 20-percent solution of trisodium cobaltinitrite. The trisodium cobaltinitrite used in the analysis shall be a blend representing each bottle in the lot. The recovery shall be within ±3 percent of the theoretical.

F–4e (2) b. *Procedure.*—Weigh a 2-gm. sample into a tared 90-mm. evaporating dish, add 10 ml. of 70-percent HNO_3 and place on the steam bath until the reaction starts. Turn off the steam until the reaction subsides, then evaporate to a gummy mass. (Do not take to dryness on a hot plate.) Place the dish on a nichrome triangle on a tripod and heat cautiously from the top by means of a burner. Finally heat from the bottom until the bulk of the carbonaceous matter is removed, then transfer to a muffle furnace at dull red heat for 10 to 15 minutes. A small amount of carbon can be neglected. Allow the dish to cool, rinse the sides with 5 ml. of water, then add 5 ml. of concentrated HCl. Cover the dish with a watch glass, place the dish on the steam bath for 5 to 10 minutes, dilute the contents of the dish to approximately 25 ml., filter the solutions through a filter paper into a 150 ml. beaker, and wash the residues thoroughly. Evaporate the filtrate to dryness on the steam bath. Add 25 ml. of 1 N HNO_3, and cool to 20° C. By means of a graduate, add 15 ml. of cooled (20° C.) 1 N trisodium cobaltinitrite solution (see par. F–4e (2) a) with stirring, to the cooled solution containing potassium ions. Cover the resulting mixture with a watch glass and allow to stand for 2 hours at 20° C. Filter the solution through a tared filtering crucible. Use a minimum amount of 0.01 normal nitric acid solution in a wash bottle to transfer the precipitate of dipotassium-sodium cobaltinitrite from the beaker to the crucible. Wash the precipitate with three 10-ml. portions of 0.01 normal nitric acid solution and then with three 5-ml. portions of alcohol. Aspirate the crucible until the contents are fairly dry, place the crucible and contents in an oven at 100° C. for 1 hour, cool in a desiccator and weigh.

$$\text{Percent potassium sulfate} = \frac{38.4A}{W}$$

where

A=gm. of precipitate.
W=gm. of the volatiles-free sample.

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F–4e (3). *Tin.*—Transfer an accurately weighed portion of approximately 2 gm. of the powder to a 300-ml. flask equipped with a side arm. Add 5 ml. of concentrated H_2SO_4 and 5 ml. of concentrated HNO_3. Heat on a hot plate to decompose the powder and continue heating to the evolution of SO_3 fumes. Cool, add 5 ml. of concentrated HNO_3 and again heat to the evolution of SO_3. Repeat the addition of HNO_3 and heating until the organic matter is destroyed and the solution is clear and colorless after cooling with the exception of a little graphite. Add 2 ml. of water and again heat to SO_3 fumes. Repeat the addition of 2 ml. of water and heating to insure the complete removal of HNO_3. Cool. Add 100 ml. of water, 10 ml. of concentrated HCl and filter through ashless paper. Retain the filter paper containing the insoluble residue for the ash determination specified in paragraph F–4e (4). Add 2 strips of nickel (approximately 1 square inch in area by 0.020 inch nominal thickness) and heat the solution at such a rate that a copious stream of fine bubbles rises from the strips, but no appreciable boiling occurs. Reduce the solution for 45 minutes and then start passing a fine stream of CO_2 into the flask by means of a gooseneck, one end of which extends into a 10-percent $NaHCO_3$ solution. On cooling, some $NaHCO_3$ is drawn into the flask with the resultant evolution of CO_2. Cool the flask to approximately 20° C. and titrate with 0.02 *N* iodine solution using starch solution indicator. Calculate the percent tin as follows:

$$\text{Percent tin} = \frac{5.935\ VN}{W}$$

where

V=ml. iodine solution used in titration
N=normality of iodine solution used
W=gm. of sample (volatiles-free)

F–4e (4). *Ash.*—Ignite two 50-ml. porcelain crucibles and lids and cool in a desiccator. Designate the pairs of crucibles and lids as assemblies A and B. Weigh assembly A using assembly B as a counterpoise. Transfer a weighed portion of approximately 5 gm. of powder to assembly A and add 5 ml. of concentrated HNO_3 to each crucible. Place the assemblies on a steam bath, adjust the lids to permit the escape of volatile matter, and heat until the powder has been converted to a gummy mass. Transfer the assemblies to a hot plate (approximately 250° C.) and drive off volatile matter. Place in an electric muffle furnace at 600° C. and ignite for approximately 2 hours to remove carbonaceous matter. Cool and add 5 drops of concentrated H_2SO_4 to each crucible. Return to the hot plate and heat till fuming ceases. Transfer to the muffle furnace, ignite for 15 minutes, remove and cool in a desiccator. Weigh assembly A using assembly B as the counterpoise. Calculate the percent ash as follows:

$$\text{Percent ash} = \frac{100A}{W} - B - 1.27C$$

where

A=increase in weight of assembly *A*
B=percent potassium sulfate. (See par. F–4e (2).)
C=percent tin. (See par. F–4e (3).)
W=gm. of sample.

F–4e (5). *Nitrocellulose.*—The percent nitrocellulose shall be calculated by subtracting from 100 percent, the sum of the percentage of

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dinitrotoluene, diphenylamine, potassium sulfate, tin, ash, and graphite. The graphite percentage which is usually very small shall be furnished by the contractor on the basis of the manufacturing data.

F–4f. *Stability test at 134.5° C.*—Five samples (each from a different chemical sample) of 2.5 gm. each shall be weighed out and placed in heavy glass tubes, 290 mm. long, 18 mm. outside diameter, and about 15 ml. inside diameter. A piece of standard normal methyl violet paper shall be placed in each tube with the lower edge 25 mm. above the top of the powder. The tubes shall be closed with cork stoppers through which a hole 4 mm. in diameter has been bored and then placed in a constant temperature bath, the temperature of which shall be carefully regulated at 134.5° C. Not more than 6 or 7 mm. of the tubes shall project from the bath. The time shall be noted when the tubes are placed in the bath and 20 minutes later each tube shall be examined by withdrawing one-half its length from the bath or until the top of the paper can be seen distinctly, and replacing quickly. Thereafter, examination shall be made at 5-minute intervals only, in the same manner. The test shall be considered as completed when the paper is completely changed to a salmon pink, and the minimum test of the number of samples tested shall be taken as the test of the lot of powder. It is important that readings be made at 5-minute intervals only. If the color changes are not completed at the end of the prescribed period, although the actual turning of the methyl violet paper to complete salmon pink takes place at one minute only after the prescribed period, the time of the test is recorded as the prescribed period, plus 5 minutes.

F–4g. *Ballistic test.*—Fire a series of 30 rounds in a velocity and pressure gage, at a weight of charge sufficient to give the prescribed velocity of 2,780 f./s. at 90 feet (corrected) with the 2,000 grain M97 Projectile, recording the velocity and pressure of each round. Use powder from a different box sample on each of the first 10 rounds of the series repeating on the second 10 rounds. Compute the average pressure (corrected to the 2,780 f./s. level) the maximum dispersion in velocity and the average deviation from the mean.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT.

G–1. *Packing.*—Unless otherwise specified, the powder shall be packed in accordance with the applicable Ordnance Department Drawing 76–4–46 or 76–4–56. All containers, before and after packing, shall be tested by the contractor and only watertight containers shall be used. Containers complying with drawing 76–4–46 shall be tested with 2–3 p. s. i. air pressure, whereas those complying with drawing 76–4–56 shall be tested with ½–1 p. s. i. air pressure.

G–2. *Marking.*—In addition to any special marking required by the contract or order, and marking to insure safe handling as required by Interstate Commerce Commission Regulation for Transportation of Explosives and Other Dangerous Articles, etc., shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES.

H–1. *Use.*—The propellent powder covered by this specification is intended for use as a propelling charge for 20-mm. guns.

H–2. Requests, requisitions, schedules, and contracts or orders should specify the following features:

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(a) Title, number, and date of the specification.
(b) Place of test of ballistic sample. (See par. F–2a.)

H–3. In case the powder fails to pass the ballistic test in the weapon for which it is intended, the bureau or agency concerned may, if deemed to its interest, accept the powder, upon its meeting the specifications for propellent powder for any other weapon.

H–4. Copies of Army Ordnance Drawings may be obtained from the Office, Chief of Ordnance, Department of the Army, Washington 25, D. C.

H–5. Information as to the availability of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., may be obtained upon application to the Interstate Commerce Commission, Washington 25, D. C.

H–6. Copies of Joint Army-Navy specifications (required for Army and Air Force purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the Departments of the Army and Air Force will obtain copies of Joint Army-Navy, United States Army, and Federal specifications through established departmental channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–7. Copies of Joint Army-Navy specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Norfolk, Va. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–8. Copies of this Joint Army-Navy specification (required for Army purchases) may be obtained from the Office, Chief of Ordnance, Department of the Army, Washington 25, D. C.

H–9. Unless otherwise specified herein, all reference to Army includes the Air Force.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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